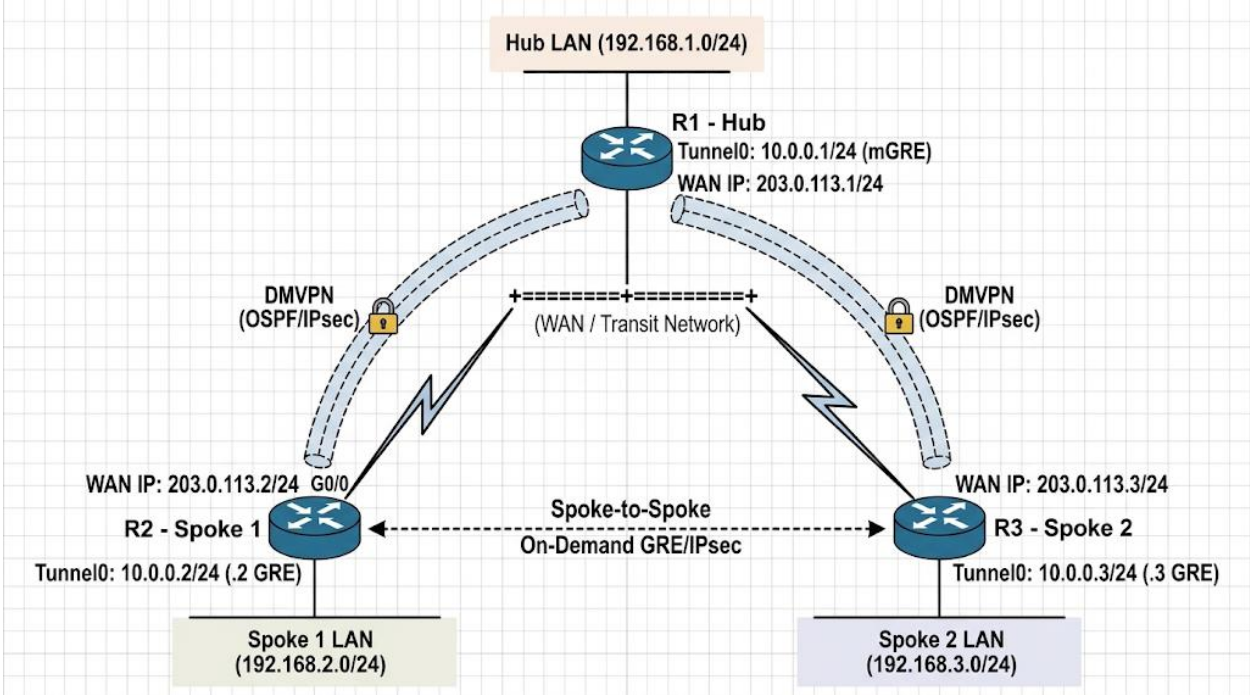
Comprehensive DMVPN Lab Guide (OSPF Version)

Designed for CCNP Students.

Topology Diagram (Logical)



Lab 1: DMVPN Phase 1 (Hub-and-Spoke Only)

OSPF Note: When running OSPF over DMVPN, we must change the OSPF network type. We will use `broadcast` and force the Hub to be the Designated Router (DR) by setting its priority to 255, and the spokes to 0.

1. Hub (R1) Configuration

```
interface Tunnel0

 description DMVPN-Hub

 ip address 10.0.0.1 255.255.255.0

 tunnel source GigabitEthernet0/0

 tunnel mode gre multipoint
```

```
 ! NHRP Configuration
 ip nhrp network-id 1
 ip nhrp map multicast dynamic
 ! OSPF Configuration
 ip ospf network broadcast
 ip ospf priority 255
 no shutdown
 exit
router ospf 1
 router-id 1.1.1.1
 network 10.0.0.0 0.0.0.255 area 0
 network 192.168.1.0 0.0.0.255 area 0
 exit
```

2. Spoke 1 (R2) Configuration

```
interface Tunnel0
 description DMVPN-Spoke1
 ip address 10.0.0.2 255.255.255.0
 tunnel source GigabitEthernet0/0
 tunnel destination 203.0.113.1
 ! NHRP Configuration
 ip nhrp network-id 1
 ip nhrp map 10.0.0.1 203.0.113.1
```

```
 ip nhrp map multicast 203.0.113.1

 ip nhrp nhs 10.0.0.1

 ! OSPF Configuration

 ip ospf network broadcast

 ip ospf priority 0

 no shutdown

 exit

router ospf 1

 router-id 2.2.2.2

 network 10.0.0.0 0.0.0.255 area 0

 network 192.168.2.0 0.0.0.255 area 0

 exit
```

(Spoke 2 uses the exact same configuration as Spoke 1, changing its Tunnel IP to 10.0.0.3, router-id to 3.3.3.3, and LAN to 192.168.3.0).

Lab 2: DMVPN Phase 2 (Dynamic Spoke-to-Spoke)

1. Hub (R1) Changes

Note: Unlike EIGRP, OSPF does not use split-horizon or next-hop-self in the same way, so no routing protocol changes are needed on the Hub for OSPF in Phase 2.

```
! No Hub changes needed for OSPF routing in Phase 2!
```

2. Spoke 1 & Spoke 2 Changes

```
interface Tunnel0

 ! Remove the hardcoded destination
```

no tunnel destination 203.0.113.1

! Convert to Multipoint GRE

tunnel mode gre multipoint

exit

Lab 3: DMVPN Phase 3 (Hierarchical & Highly Scalable)

1. Hub (R1) Changes

interface Tunnel0

! Enable NHRP Redirects on the Hub

ip nhrp redirect

exit

2. Spoke 1 & Spoke 2 Changes

interface Tunnel0

! Allow Spokes to install direct shortcut routes

ip nhrp shortcut

Exit

Full configuration

R1 (Hub) Complete Configuration

```
configure terminal

!

interface GigabitEthernet0/0

 description WAN-Link
```

```
 ip address 203.0.113.1 255.255.255.0
 no shutdown
 exit
!
interface GigabitEthernet0/1
 description Hub-LAN
 ip address 192.168.1.1 255.255.255.0
 no shutdown
 exit
!
interface Tunnel0
 description DMVPN-Hub-Phase3
 ip address 10.0.0.1 255.255.255.0
 tunnel source GigabitEthernet0/0
 tunnel mode gre multipoint
 ip nhrp network-id 1
 ip nhrp map multicast dynamic
 ip nhrp redirect
 ip ospf network broadcast
 ip ospf priority 255
```

```
 no shutdown
 exit
!
router ospf 1
 router-id 1.1.1.1
 network 10.0.0.0 0.0.0.255 area 0
 network 192.168.1.0 0.0.0.255 area 0
 exit
!
end
write memory
```

R2 (Spoke 1) Complete Configuration

```
configure terminal
!
interface GigabitEthernet0/0
 description WAN-Link
 ip address 203.0.113.2 255.255.255.0
 no shutdown
 exit
!
```

```
interface GigabitEthernet0/1
 description Spoke1-LAN
 ip address 192.168.2.1 255.255.255.0
 no shutdown
 exit
!
interface Tunnel0
 description DMVPN-Spoke1-Phase3
 ip address 10.0.0.2 255.255.255.0
 tunnel source GigabitEthernet0/0
 tunnel mode gre multipoint
 ip nhrp network-id 1
 ip nhrp map 10.0.0.1 203.0.113.1
 ip nhrp map multicast 203.0.113.1
 ip nhrp nhs 10.0.0.1
 ip nhrp shortcut
 ip ospf network broadcast
 ip ospf priority 0
 no shutdown
 exit
!
```

```
router ospf 1
 router-id 2.2.2.2
 network 10.0.0.0 0.0.0.255 area 0
 network 192.168.2.0 0.0.0.255 area 0
 exit
!
end
write memory
```

R3 (Spoke 2) Complete Configuration

```
configure terminal
!
interface GigabitEthernet0/0
 description WAN-Link
 ip address 203.0.113.3 255.255.255.0
 no shutdown
 exit
!
interface GigabitEthernet0/1
 description Spoke2-LAN
 ip address 192.168.3.1 255.255.255.0
```

```
 no shutdown
 exit
!
interface Tunnel0
 description DMVPN-Spoke2-Phase3
 ip address 10.0.0.3 255.255.255.0
 tunnel source GigabitEthernet0/0
 tunnel mode gre multipoint
 ip nhrp network-id 1
 ip nhrp map 10.0.0.1 203.0.113.1
 ip nhrp map multicast 203.0.113.1
 ip nhrp nhs 10.0.0.1
 ip nhrp shortcut
 ip ospf network broadcast
 ip ospf priority 0
 no shutdown
 exit
!
router ospf 1
 router-id 3.3.3.3
```

```
 network 10.0.0.0 0.0.0.255 area 0
 network 192.168.3.0 0.0.0.255 area 0
 exit
!
end
write memory
```